SUSHANTH / PAATNAIK

INVENTOR • DEEP-TECH • FOUNDER

Engineering Matter. Capital. Scale.

Sushanth Paatnaik is an inventor, deep-tech entrepreneur, and industrial systems architect who has spent two decades engineering materials that redefine what is possible at the frontier of science and industry. Holder of 40+ international patents, Presidential awardee, MIT TR35 honoree, and TED speaker — he transforms laboratory breakthroughs into ventures that reshape entire industries. His companies operate at the intersection of advanced materials, clean energy, and infrastructure resilience, collectively raising over ₹200 crore in capital and deploying technology at national scale.

ORIGIN

*"Every idea has a **birthplace.**"*

- Bhubaneswar, Odisha
- Started inventing at 17
- Materials science pioneer

JOURNEY

EARLY INVENTION
The garage workshop

NATIONAL RECOGNITION
Presidential Award 2014

TED SPEAKER
Ideas Worth Building

GLOBAL PATENTS
40+ Protected Inventions

DEEP-TECH VENTURES
Building Companies

R&D EXPANSION
Lab to Industry Scale

FUTURE SYSTEMS
The mission continues

PHILOSOPHY

"The future is not imagined. It is engineered quietly."

Sushanth believes that true innovation is never loud. It happens in laboratories at midnight, in material characterization runs, in field deployments that survive monsoons and decades. His philosophy: engineer the atoms first, and the future follows.

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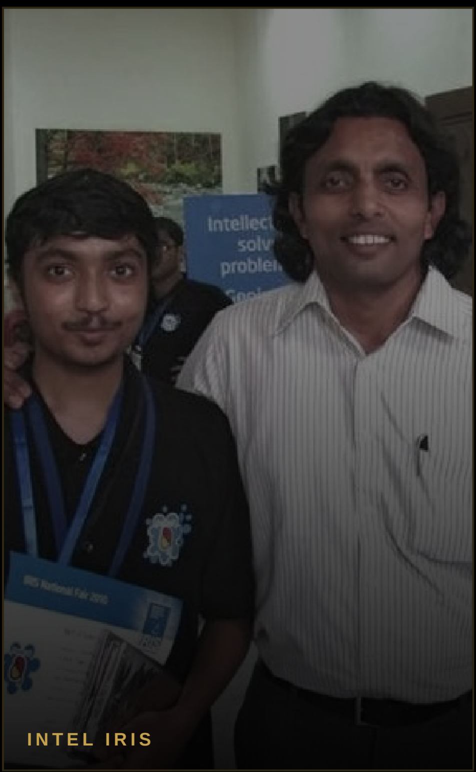
HONORS & RECOGNITIONS

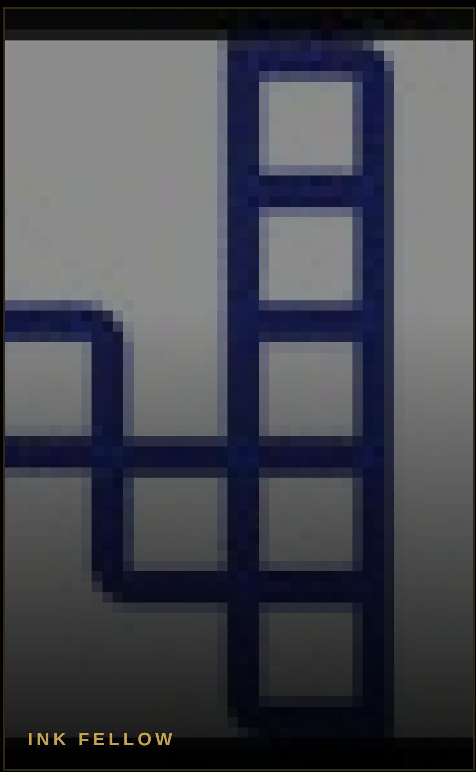

PRESIDENTIAL AWARD

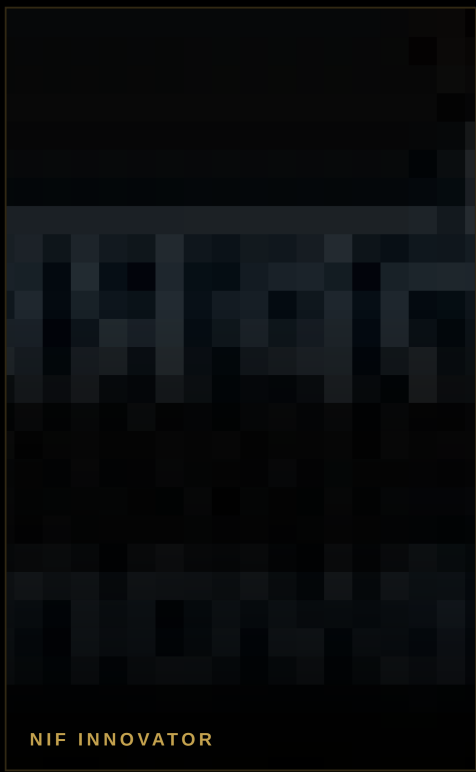

NASA INNOVATION


MIT TR35

TED SPEAKER


INTEL IRIS


INK FELLOW


NIF INNOVATOR

BY THE NUMBERS

40+ PATENTS FILED WORLDWIDE	**6** DEEP-TECH VENTURES
₹200Cr+ CAPITAL RAISED	**15+** YEARS OF INNOVATION
50+ MEDIA FEATURES	**3** PRESIDENTIAL HONORS

ENGINEERING INTELLIGENT MATTER

ADVANCED MATERIAL SYSTEMS ENGINEERED FOR INDUSTRIAL PERFORMANCE

GRAPHACRETE

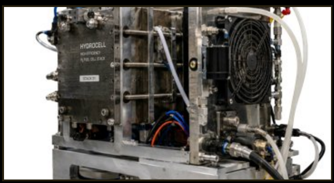
Graphene-enhanced concrete composite delivering 3× compressive strength with 40% reduced carbon footprint. Deployed in critical infrastructure across India.

GRAFFISOL

Graphene-based soil stabilization system improving load-bearing capacity by 5× for road and railway foundation applications in challenging terrain.

IGNITRON-D

Next-generation defense-grade energetic material with precise detonation control. Developed for strategic applications with national security agencies.



HYDROCELL

Nano-engineered water purification membrane achieving 99.9% heavy metal removal at scale. Deployed across municipal water systems.

AQUAMAX

Advanced hydrophobic surface treatment reducing water absorption in construction materials by 95%, extending structural life in humid climates.

INNOVATION PORTFOLIO

ARMOPHENE
LIGHTWEIGHT ARMOR COMPOSITE

• • • •

32%
WEIGHT REDUCTION
Graphene-reinforced armor panels for defense and security applications

CERAPHENE
ULTRA-HIGH TEMP CERAMIC

• • • •

1800°
THERMAL RESISTANCE
Ceramic-graphene composite for extreme-temperature industrial environments

VOLTAPHENE
CONDUCTIVE COMPOSITE

• • • •

3×
CONDUCTIVITY
High-performance electrode material for next-gen energy storage systems

LUBRITRON
NANO LUBRICANT SYSTEM

• • • •

80%
FRICTION REDUCTION
Graphene-oil nano-dispersion extending component life in heavy machinery

COALORIX
CLEAN COAL ADDITIVE

• • • •

60%
EMISSION CUT
Nano-catalyst additive reducing particulate and SO_2 emissions from thermal plants

PATENTED | LAB TESTED | FIELD DEPLOYED

INDUSTRIAL TRANSLATION

From advanced materials lab to industrial deployment — six sectors transformed by intelligent matter.

SOLAR	ENERGY	AUTOMOTIVE
INFRASTRUCTURE	DEFENSE	CONSTRUCTION

VENTURES

GT

GrafTech Industries
Advanced Materials Manufacturing

Building the production infrastructure for graphene-composite materials at industrial scale across infrastructure and defense verticals.

SP

SolarPher Energy
Next-Gen Solar Technology

Deploying graphene-enhanced photovoltaic systems delivering 22% higher efficiency than conventional panels for utility-scale solar farms.

AP

AquaPure Systems
Advanced Water Purification

Nano-membrane water treatment solutions serving municipalities and industrial facilities across South and Southeast Asia.

DM

DefenseMat Corp
Strategic Defense Materials

Classified-grade advanced materials for defense applications including armor, energetics, and structural systems for Indian armed forces.

SC

SmartCrete Solutions
Intelligent Construction Materials

Graphacrete and Graffisol deployment for national highway projects, smart cities infrastructure, and railway expansion programmes.

ES

EnergyStore Labs
Energy Storage Innovation

Next-generation supercapacitor and battery systems using Voltaphene electrodes for grid-scale and mobility energy storage applications.

FUNDED | OPERATIONAL | SCALING

GLOBAL RECOGNITION

THE HINDU

"India's graphene pioneer is quietly building tomorrow's infrastructure"

March 2023

FORBES INDIA

"30 Under 30: The inventor making India's roads smarter"

2019

ECONOMIC TIMES

"From Odisha garage to global patent portfolio — a deep-tech journey"

Jan 2022

ENTREPRENEUR

"How Sushanth Paatnaik is engineering the future of Indian industry"

Aug 2021

BUSINESSLINE

"₹200 crore raised for graphene ventures disrupting construction and defense"

Nov 2022

MINT

"Presidential awardee builds deep-tech empire on the science of atoms"

Feb 2023

NDTV

"Meet the inventor whose material science is protecting Indian soldiers"

July 2022

DECCAN HERALD

"NIF innovator turns graphene into a platform for national development"

April 2021

FEATURED IN 50+ NATIONAL & GLOBAL MEDIA

SPEAKING ENGAGEMENTS

SHARING IDEAS AT THE WORLD'S MOST IMPORTANT STAGES

TED / TEDx Stage

Delivered a talk on engineering intelligent matter and how graphene composites will redefine industrial civilization. Viewed by over 2 million people globally.

CEO Clubs — Silicon Valley

Keynote on deep-tech entrepreneurship and capital formation strategies for frontier technology companies building from emerging markets.

BRICS Innovation Forum

Represented India's deep-tech ecosystem at the BRICS summit on shared research infrastructure and cross-border technology collaboration.

G20 Innovation Summit

Panelist on sustainable industrial materials and the role of nano-science in achieving net-zero targets across G20 member economies.

IEEMA Industry Summit

Addressed India's electrical and electronics manufacturing association on the integration of advanced materials in next-generation power infrastructure.

FUTURE SYSTEMS

MATERIALS ENERGY INDUSTRY WATER HYDROGEN AI

"

The next industrial revolution will be built on intelligent materials, clean energy, and frontier technologies.

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COLLABORATION

INDUSTRY	INVESTMENT	RESEARCH	GOVERNMENT
Co-creating impact at scale through joint ventures and technology licensing partnerships	Funding deep-tech moonshots that solve civilization-scale problems through patient capital	Pushing the boundaries of science through academic partnerships and lab-to-market translation	Building national systems for the future through policy engagement and strategic programmes

Let's build the future together.

SUSHANTHPAATNAIK.COM

"Building industrial systems that outlive the inventor."

SUSHANTH PAATNAIK

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